

MEREDITH STONE

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EDUCATION

University of Arizona

2022 - present

Ph.D. candidate; advanced to candidacy May 2024

NSF Graduate Research Fellow

University of Massachusetts Amherst

2018 - 2022

B.S. Astronomy and Physics

Summa cum laude; Commonwealth Honors College Scholar with greatest distinction

RESEARCH EXPERIENCE

University of Arizona

Tucson, AZ

Department of Astronomy/Steward Observatory

2022 - present

Advisors: Professor George Rieke, Professor Stacey Alberts

- Probing the host galaxies of high-redshift quasars with JWST/NIRCam photometry, using sophisticated image deconvolution techniques to investigate the coevolution of galaxies and their supermassive black holes over cosmic time
- Comparing predictions from galaxy formation models and existing infrared data to new, unprecedentedly deep number counts of galaxies in JWST/MIRI photometric surveys
- Constraining an overdensity of galaxies around a high-redshift quasar with narrow-band selection from deep JWST/NIRCam images
- Exploring the evolution of interstellar medium conditions out to cosmic noon with JWST/MIRI low-resolution spectroscopy of mid-infrared dust features

University of Massachusetts Amherst

Amherst, MA

Five College Astronomy Department

2019 - 2022

Advisor: Professor Alexandra Pope

- Performed a novel stacking analysis on Spitzer/IRS mid-infrared spectra of (ultra)luminous infrared galaxies
- Developed a correction to a key spectral tracer of black hole accretion for contamination from star formation, allowing for more robust black hole accretion measurements

Yale University

New Haven, CT

Department of Astronomy

2021

Advisor: Professor C. Megan Urry

- Awarded Dorrit Hoffleit Undergraduate Research Scholarship to pursue a summer of research at Yale
- Created a simulated sample of galaxy and active galactic nucleus (AGN) photometry for use in training a deep learning model to separate normal galaxies from AGN
- Reproduced the optical and infrared colors of quiescent and star-forming galaxies and Type I and Type II AGN

PUBLICATIONS

Stone, M., Rieke, G., Lyu, J., et al. “The $z = 7.08$ Quasar ULAS J1120+0641 May Never Reach a “Normal” Black Hole to Stellar Mass Ratio.” Accepted to *The Astrophysical Journal*, September 2025; preprint available to read [on arxiv.org](#).

Lyu, J., Rieke, G., **Stone, M.,** et al. “Fading Light, Fierce Winds: JWST Snapshot of a Sub-Eddington Quasar at Cosmic Dawn.” *The Astrophysical Journal* 981.20, March 2025. [Available here](#).

- Stone, M.,** Alberts, S., Rieke, G., et al. “5-25 μm Galaxy Number Counts from Deep JWST Data.” *The Astrophysical Journal* 972.62, August 2024. [Available here.](#)
- Stone, M.,** Lyu, J., Rieke, G., Alberts, S., and Hainline, K. “Undermassive Host Galaxies of Five $z \sim 6$ Luminous Quasars Detected with JWST.” *The Astrophysical Journal* 964.90, March 2024. [Available here.](#)
- Stone, M.,** Lyu, J., Rieke, G., and Alberts, S. “Detection of the Low-Stellar Mass Host Galaxy of a $z \sim 6.25$ Quasar with JWST.” *The Astrophysical Journal* 953.180, July 2023. [Available here.](#)
- Stone, M.,** Pope, A., McKinney, J., et al. “Measuring Star Formation and Black Hole Accretion Rates in Tandem using Mid-Infrared Spectra of Local Infrared-Luminous Galaxies.” *The Astrophysical Journal* 934.27, July 2022. [Available here.](#)

PRESENTATIONS AND OTHER WRITING

- “The Sparse Environment of the $z = 7.1$ Quasar ULAS J1120+0641.” Oral presentation at 2025 meeting of the NIRCам instrument team, March 2025.
- “Slit-Stepping for PAHs with MIRI Low-Resolution Spectroscopy.” Oral presentation, Building Physical Simulations of PAH Emission in Galaxies workshop, November 2024.
- “Galaxy Counts Reach New Depths with MIRI.” Oral presentation at 2024 meeting of the MIRI instrument team, October 2024.
- “Constraining the Stellar Masses of High L_{AGN} $z \sim 5 - 7$ Quasars.” Oral presentation at meeting of the EREBUS collaboration, October 2023.
- “The Black Hole Mass-Stellar Mass Relation in $z \sim 6$ Quasars with NIRCам.” Poster presentation, STScI’s First Year of JWST Science conference, September 2023.
- “Spotting the Difference: Supermassive Black Holes in Cosmological Simulations Don’t Match Observations.” [Astrobites.org](#), May 2023.
- “Measuring Star Formation Rates and Black Hole Accretion Rates in Tandem using Mid-Infrared Spectra of Local Infrared-Luminous Galaxies.” Poster presentation, 240th meeting of the American Astronomical Society, June 2022. [Available to view here.](#)
- “The Coevolution of Star Formation and AGN Accretion in Local Infrared-Luminous Galaxies.” Oral presentation, New England Regional QUasar and AGN Meeting (NERQUAM), May 2022.
- “UR: Understanding the Balance of Star Formation and Black Holes in Nearby Luminous Infrared Galaxies.” [Astrobites.org](#), March 2021.
- “Uncovering the Relationship between Stars and Black Holes in Nearby Luminous Infrared Galaxies: Probing Lines Below the Noise with Stacked *Spitzer*/IRS Spectra.” Virtual poster presentation, 237th meeting of the American Astronomical Society, January 2021. [Available to view here.](#)

GRANTS AND AWARDS

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| Graduate Student Award in Service
<i>Steward Observatory/Department of Astronomy</i> | 2025 |
| NSF Graduate Research Fellowship
<i>National Science Foundation</i> | 2024 |
| NSF Graduate Research Fellowship - Honorable Mention
<i>National Science Foundation</i> | 2022 |
| Mary Dailey Irvine Thesis Prize
<i>Five College Astronomy Department</i> | 2022 |

Department of Astronomy Outstanding Senior Award <i>UMass Amherst Department of Astronomy</i>	2022
Philip Hasbrouck Scholarship for Academic Excellence <i>UMass Amherst Department of Physics</i>	2022
Goldwater Scholarship <i>Barry Goldwater Excellence in Education Foundation</i>	2021
Dorrit Hoffleit Undergraduate Research Scholarship <i>Yale University Department of Astronomy</i>	2021
Chambliss Astronomy Achievement Student Award - Honorable Mention <i>American Astronomical Society</i>	2021
Commonwealth Honors College Research Assistant Fellowship <i>UMass Amherst Commonwealth Honors College</i>	2020, 2021
Massachusetts Space Grant Consortium In-State Fellowship <i>Massachusetts Space Grant Consortium</i>	2019, 2020, 2021
David J. Van Blerkom Research Scholarship <i>Five College Astronomy Department</i>	2019

TEACHING AND MENTORING EXPERIENCE

Graduate Coordinator, TIMESTEP Research Apprenticeship 2023 - present

- Mentored undergraduate students undertaking first research projects through the [TIMESTEP](#) program
- Developed and led research skills workshops (reading scientific papers, using NASA ADS, Python, $\text{\LaTeX}$, scientific writing, and oral and poster presentations)
- Interfaced with TIMESTEP program leaders, other graduate coordinators, and faculty research advisors to assist students and facilitate positive research experiences
- Created a [research map](#) of University of Arizona space sciences faculty as an undergraduate student resource

Teaching Assistant, Steward Observatory 2024

- Graduate TA for cross-listed graduate/undergraduate instrumentation course
- Obtained certified observer status on University of Arizona 21" and 61" telescopes, and led observing runs with student groups
- Developed and delivered lecture on photometry methods

Teaching Assistant, UMass Amherst Physics 2019 - 2022

- Worked with small groups of students (non-physics majors, introductory physics sequence) in a team-based learning environment to facilitate discussion and answer questions
- Assisted students in a wide variety of physics classes in the Physics Help Room
- Hosted virtual office hours and review sessions during remote semesters

OUTREACH AND PROFESSIONAL EXPERIENCE

Staff Member, Arizona Astronomy Camp 2024 - present

- Led activities exploring the behavior of light and astronomical data reduction techniques (images and spectra), and assisted campers in independent research projects
- Operated Arizona research-class telescopes for eyepiece and CCD observing
- Completed training to interact with non-enrolled minors and to drive University of Arizona vehicles
- Maintained the [2024 Advanced Teen Camp blog](#) to update parents on camp activities

Steward Observatory Graduate Council*2023 - present*

- Served as a representative of Astronomy graduate students in interfacing with department leadership
- Aided incoming first-year graduate students over the summer in the process of moving to Tucson: facilitated apartment tours, trips to pick up furniture, and welcome events
- With council members, facilitated merchandise sales, ran the department coffee machine, and managed the revenue to host graduate student events

Judge, Southern Arizona Regional Science and Engineering Fair*2023 - present***Graduate Representative, Academic Program Review Committee***2024 - 2025***Graduate Coordinator, TIMESTEP Summer Tech Internship***2024*

- Mentored five undergraduate STEM students pursuing technical summer internships in the Tucson area
- Supported students through the process of creating an appropriate resume and cover letter; opened communication between students and company matches
- Met weekly with students to check in and offer professional guidance
- Worked with other coordinators to offer professional development opportunities to the interns

Five College Astronomy Club*2018 - 2022*

- Secretary, 2019 - 2020; President, 2020 - 2022
- Circulated a weekly astronomy newsletter to over 100 students during remote semesters, and received highly positive feedback from students and faculty
- Coordinated speakers, panels, and special topics lectures to appeal to both astronomy students and the general Five College population